

Model for Molten Salt Absorption Coefficient Using the Pair Distribution Function

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INTRODUCTION

Nuclear power provides an alternative energy source to fossil fuels as global power demand increases and sustainable energy sources are pursued. The NRC requires extensive parameterization of phenomena before construction of novel reactor designs, which has led to many engineering questions in the development of fourth generation nuclear reactors. As an example, Kairos Power's Fluoride Salt-Cooled High-Temperature Reactor (FHR) involves the flow of fluoride molten salt through a bed of TRISO pebbles [1], which presents the challenge of ensuring adequate accounting for contributions to heat transfer in the molten salt coolant. Investigations of MSR heat transfer have been generally limited to convection [2]. Incorporating radiation into heat transfer models of the molten salt-pebble system will provide a more complete picture of the heat transfer behavior in an FHR core.

Using data from constituent salts to estimate the absorption coefficient, κ , radiative heat transfer (RHT) has previously been estimated for FLiNaK (eutectic LiF-NaF-KF) in laminar flow in a circular pipe [3]. At 973K, (target core outlet temperature of the Mk1 PB-FHR) Chaleff found RHT to account for 10% of total heat transfer at a pipe diameter of about 2.4 cm, with larger diameters allowing more absorption, leading to a greater contribution from RHT. Moreover, conflicting studies of the transmissivity of molten salt mixtures disagree with conservative estimates used by Chaleff [4]. Despite the potential impact to MSR predictability and control, the lack of data constrains composition-dependent parameterization of optical properties.

Molten salt heat transfer models tend to use gray approximations due to the limited availability of data for higher-order mixtures, owing to the experimental challenges of molten salts and the still limited understanding of thermophysical property data [5]. Laser-based thermal conductivity measurements such as laser flash analysis (LFA) and transient gradient spectroscopy (TGS) can be especially impacted [6, 7], where the window and molten salt absorption coefficient can have pronounced influence under high-energy lasers [8]. These challenges stress the need for increased

The lack of spectroscopic data has been addressed in recent decades using molecular dynamic (MD) simulations [9] to supplement extensive reviews of what available data exists [8], [10], though the variety of molten salt compositions makes the tabulation process a limiting factor to the optimization and safety prediction of molten salts for MSRs. Additionally, the temperature ranges involved show non-negligible changes in absorption coefficient for a range of salts, suggesting the need for accurate temperature-dependent relationships [11, 12, 13].

The molecular structure of molten salts is closely related to its absorptivity. Because molten salt operating temperatures generally lie within the mid-to-far-infrared range corresponding to vibrational quantum energy states, structural

variations in cation-anion and interionic complexes can have complex effects on their absorption of thermal radiation [11]. Ab initio molecular dynamics (AIMD) can begin to piece together the quantum picture involved in photon absorption, but these results are time-intensive and lack a direct interpretation that yields the absorptivity of molten salts. To elucidate a generalizable relationship between constituents of molecular structures and bulk absorptance, a more direct connection needs to be made between vibrational modes of molten salt structures and the resulting absorptivity.

The purpose of this study is to develop a model of the temperature-dependent absorption coefficient in terms of molten salt molecular structure. Comparisons of absorptivity predictions to the limited data available will be used to verify phenomenological trends in the compositional dependency of radiative absorption. The results of this work can provide insights for further development of a compositionally-dependent predictive model of the absorption coefficient informed by first-principles.

MODELING APPROACH

Using material properties of graphite at 700C, the emissions will be calculated leaving the pebble (accounting for the spectral distribution of the power). Using Beer's law and our estimated absorptive coefficient, the power absorbed by the salt will be calculated in a simple one dimensional model and compared against convection heat transfer (based on Nusselt number correlations from literature [4]). If found to be significant, modeling will begin in ANSYS Fluent, using the discrete ordinates method. A sensitivity analysis and comparison of results for measured and predicted absorption coefficients will be conducted.

Absorption Coefficient Modeling

To estimate the spectral absorption coefficient of molten salts as a function of the constituent salt composition, a predictive model must relate the probability of radiative absorption to the energy storage modes of the system. Fundamentally, the spectral absorption coefficient in a material is proportional to the probability of transition between energy states after photon interaction, given by:

$$\kappa(\omega) = \sum W(i \rightarrow f) n_i P_f \quad (1)$$

where $\kappa(\omega)$ is the spectral absorption coefficient for frequency ω (equivalent to κ_λ for wavelength λ), $W(i \rightarrow f)$ is the transition probability between initial state i and final state f , n_i is the occupation number of the quantum state, and P_f is the probability of an open final state ($= 1$ for bosons like vibrations). Eq. (1) can be expressed in terms of the density of states, $D(E)$, and relative availability for optical absorption

across the energy available in the range $0 < E < \infty$:

$$\kappa(\omega, T) = \int_0^\infty D(E)D(E + \hbar\omega)e^{-E/k_B T} dE \quad (2)$$

Rather than calculating $D(E)$, which requires AIMD modeling or back-calculations from experimental values, the shape of the absorption coefficient curve can be approximated by estimating the distribution of frequency-dependent energy storage modes. Senatore et. al. [14] proposed a simple model of fundamental ultraviolet and visible absorption in molten alkali halides. Based on the semi-empirical Urbach Rule, the electronic absorption coefficient, κ_{elec} , is defined as:

$$\kappa_{\text{elec}}(\omega, T) = \kappa_0 e^{\frac{\hbar\omega - W_0}{k_B T}} \quad (3)$$

where $\hbar$ is the reduced Planck constant, T is the temperature, k_B is Boltzmann's constant, κ_0 is a fitted constant baseline absorption coefficient, and W_0 is the energy of the peak ultraviolet absorption, reported in [14].

At the working temperatures of molten salts, comparatively little of the energy emitted and absorbed falls within this ultraviolet (UV) range compared to the dominating infrared wavelengths. The absorption modeling of this work thus focuses primarily on the vibrational contribution, κ_{vib} , to the total absorptive coefficient, κ_{total} :

$$\kappa_{\text{total}}(\omega, T) = \kappa_{\text{vib}}(\omega, T) + \kappa_{\text{elec}}(\omega, T) \quad (4)$$

The vibrational contribution to the absorption coefficient can be estimated by a Lorentz oscillator model, given by:

$$\kappa_{\text{vib}}(\omega, T) = \frac{\omega}{c} \sqrt{2 \left(\sqrt{\varepsilon'^2 + \varepsilon''(\omega, T)^2} - \varepsilon' \right)} \quad (5)$$

where c is the speed of light, and ε' and ε'' come from the dielectric function

$$\varepsilon = \varepsilon' + i\varepsilon'' \quad (6)$$

where the real part ε' defines how much light is slowed in the medium, and the imaginary part ε'' describes how much energy from light is absorbed.

Using the dielectric function provides a physically-based shape that approximates the result of the density of states for vibrational modes. The value of ε depends on the vibrational modes in the system that are able to couple with electromagnetic fields. In a multi-component molten salt mixture, it is assumed that the dominant vibrational modes will exist between the cation-anion pairs that form the component parts of the mixture, considering the relative proximity and increased bond strength of cation-anion pairs in short-range ordered molten salts [15, 16, 17]. A simple model of cation-anion oscillators can be built by considering the total dielectric function [18] as a sum of n cation-anion pairs in the mixture:

$$\varepsilon(\omega, T) = \varepsilon_\infty + \sum_{j=1}^n \frac{\Delta\varepsilon_j + \omega_{0,j}^2}{(\omega_{0,j}^2 - \omega^2) - i\gamma_j(T)\omega} \quad (7)$$

where ε_∞ is the high-frequency background dielectric constant (related to the refractive index n_{refr} by $\varepsilon_\infty \approx n_{\text{refr}}^2$), $\Delta\varepsilon_j$ is the oscillator strength, $\omega_{0,j}$ is the fundamental frequency of

cation-anion pair j , and γ_j is a broadening function applied to the absorption bandwidth, with a base width $\gamma_{0,j}$ and slope γ'_j to form a function of temperature ($\gamma_j = \gamma_{0,j} + \gamma'_j \times T[K]$). The oscillator strength $\Delta\varepsilon_j$ is obtained from the difference between the baseline static dielectric constant, ε_s , and the high frequency ε_∞ ($\Delta\varepsilon = \varepsilon_s - \varepsilon_\infty$), which gives the strength of absorption due to the ions themselves (without the electron contribution).

The temperature dependency of the modeled absorption coefficient can also be affected by the decreasing fluid density with temperature, where the thermal energy of the ions increase relative to the bond strength. To add a first-order estimate of this effect, a linear factor is applied to the oscillator strength to scale its change with:

$$\Delta\varepsilon(T) = \Delta\varepsilon_0 \frac{\rho(T)}{\rho(T_{\text{fus}})} \quad (8)$$

where ρ_{fus} is the density at the melting temperature T_{fus} . The value for $\Delta\varepsilon_0$ provides a nominal oscillator strength of the solid salt crystal, as the literature generally provides the value at room temperature only. This assumes the calculated $\Delta\varepsilon_0$ to be relatively constant up to the solid-liquid transition, which provides an informed approximation due to the minimal change in solid density compared to liquids.

The oscillator strength weights the relative contributions of each cation-anion pair, and can be estimated with their molar concentrations x_i by:

$$\Delta\varepsilon_j = \frac{\varepsilon_{s,i} - \varepsilon_\infty}{\sum_i x_i (\varepsilon_{s,i} - \varepsilon_\infty)} \quad (9)$$

where $\varepsilon_{s,i}$ is the static dielectric constant, which has relatively minor influence and is assume = 5 in this work.

The fundamental frequency, $\omega_{0,j}$, is the most influential term in the model, and is calculated by:

$$\omega_0 = \sqrt{\frac{k}{\mu}} \quad (10)$$

where k is the bond constant and μ is the reduced mass of the cation-anion pair ($\mu = m_1 m_2 / (m_1 + m_2)$), for masses m_1 and m_2). If the interatomic potential, $V(r)$, is known, the bond constant can be obtained by evaluating the second derivative of $V(r)$ at the equilibrium distance r_0 :

$$k = \left. \frac{d^2 V}{dr^2} \right|_{r=r_0} \quad (11)$$

As a surrogate for the interatomic potential, the partial pair distribution function (PDF) of the molten salt can provide a direct expression of the resultant potential formed by the complex interactions within the short-range order structure. The PDF, given by $g_i(r)$ is a probability density function describing the likelihood of ion pairs being separated by a distance r . The potential of mean force (PMF) provides an averaged radially-dependent interaction potential between given pairs in the molten salt, calculated by $V_{\text{PMF}}(r) = -k_B T \ln g(r)$. Not only does this provide the interaction potential for calculating k , the minima formed at the location of

the pair's first peak corresponds to the energy potential minima, indicating the ideal equilibrium position r_0 . The calculation of $V_{PMF}(r)$ from the PDF of NaCl is shown in Fig. 1 as an example. Using $k_i = \frac{d^2}{dr^2} V_{PMF,i}(r_0)$ for each ion pair, the dielectric function can be calculated and summed for all cation-anion pairs involved.

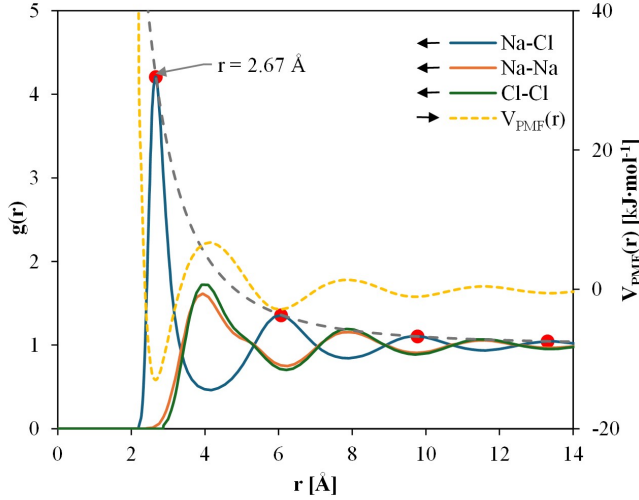


Fig. 1: PDF of NaCl with superimposed PMF function for the Na-Cl ion pair. PDF reproduced from Walz and Spoel [15].

Most model variables are obtained or extrapolated from literature for alkali and alkaline earth halides. Li et al. performed a rigorous review of spectroscopic data of alkali halide salts [18, 8], which provide ϵ_s , ϵ_∞ , n_{refr} , as well as $\gamma_{0,j}$. Senatore et al. provide values for W_0 , though κ_0 is unavailable for the range of salts studied. Where absent, the dielectric constants and Urbach are estimated from the most similar salts (noted in this work). Because the electronic contribution is not the focus of this work, all salts analyzed used a nominal value of $\kappa_0 = 1 \times 10^{12}$ to make the UV portion consistent with general alkali halide spectra [3]. Tabulated reference correlations for $\rho(T)$ and T_{fus} of unary and multi-component mixtures is provided by the Molten Salt Thermophysical Properties Database [19].

The most critical parameter absent from the available reference data is the slope of the broadening function, γ' . Li's provided values for γ_0 were often obtained at room temperature as well, limiting their applicability in molten salts. Because the fundamental frequency, dielectric constants, and other supplied model parameters provide general location of the absorption peaks, the values for γ_0 and γ' can be solved for by fitting the modeled absorption spectrum to experimental data. This is done with a Nelder-Mead algorithm whose objective function quantifies the position and slope error of the predicted versus model curves in log space, details of which are found in the attached code of Appendix ???. In multi-component mixtures, $\gamma_{0,j}$ and γ'_j are solved for each endmember j .

To study the relevance of temperature changes to the predicted absorption coefficient, a weighted temperature-dependent, wavelength-independent value was calculated for each salt. This demonstrates the bridge that make predictions

practically useful for heat transfer modeling, with applications described in Section ???. The Planck-mean absorption coefficient κ_P was obtained using wavelength (λ) averaging corresponding to the blackbody emission intensity $I_{b,\lambda}(T)$, as follows:

$$\kappa_P(T) = \frac{\int_0^\infty \kappa_\lambda I_{b,\lambda}(T) d\lambda}{\int_0^\infty I_{b,\lambda}(T) d\lambda} \quad (12)$$

Six salts were analyzed in this study: unary salts LiF, NaF, KF, and NaCl, as well as the mixtures FLiNaK and FLiBe. The modeled $\kappa_{\text{vib}}(\omega, T)$ was calculated for the working temperature range of each molten salt, starting at the melting temperature. Section provides the model parameters uses and discusses the assumptions made where data was unavailable. The associated code is attached in Appendix ??.

RESULTS AND DISCUSSION

Absorption Modeling

Table I contains the input parameters used for the absorption coefficient model for LiF, NaF, KF, NaCl, FLiNaK, and FLiBe. The PDFs of each salt was obtained from literature [15, 20, 21]. Experimental absorption coefficient data for LiF, NaCl, FLiNaK, and FLiBe [11, 22, 10] enabled the fitting of $\gamma_{0,j}$ and γ'_j for the pertinent endmembers. For all salts considered here, the measured UV and IR absorption peaks are all obtained from the room-temperature crystal. The fitted results for NaF and KF components in FLiNaK were used to supply $\gamma_{0,j}$ and γ'_j in the unary NaF and KF.

The fitted solution is compared to experimental data for NaCl in Fig. 2. The peak positions appear reasonably located in Fig. 2a compared to the experimentally obtained UV and IR peaks of the crystal. The UV absorption peak aligns predictably at very high frequencies where the salt is nearly opaque, while the maximum peak in the IR region ($\kappa_{\text{IR,model}} \approx 42.6$) lies just past the IR absorption peak of the crystal. This fits the theoretical expectation: if the bond strength decreases relative to thermal energy as the temperature increases from solid to liquid, the fundamental frequency of the cation-anion pair is likely to decrease. This is likely not due to the separation distance of the cation-anion pair, which actually tends to decrease with increasing temperature in the molten salt [7, 24, 25, 15].

While the fitted solution intersects the experimental data, the shapes do not match. This is not due to the electronic Urbach parameters, which affect only relatively small wavelengths. The fact that the optimization cannot match the experimental data shape (shown in Fig. 2b) by varying the broadening function alone may imply that the solid-state dielectric input parameters vary significantly from the molten properties, or that the PDF-derived PMF function does not capture other significant vibrational modes. If the latter is the case, additional modes are likely to increase sources of absorption, not provide transparent regions. It is also necessary to consider the relative uncertainty of the measured absorption spectra, where optical interference by windows can form one of various confounding parameters [8].

If the inferred increasing parabolic curve implied by the data is extended with increasing wavelength, the trend would

TABLE I: Input parameters for molten salt optical property modeling, where ϵ_s is the static dielectric constant, ϵ_∞ is the high-frequency dielectric constant, W_0 and κ_0 are the electronic Urbach parameters from Senatore et. al [14], $\gamma = \gamma_0 + \gamma'T$ is the vibrational damping coefficient, and $\rho(T)$ is the temperature dependent density (temperature units of [K]). Due to the lack of literature values, κ_0 is a the same nominal value used for all mixtures. Unless otherwise noted, parameters were obtained from literature or fitted to experimental data [8, 11, 23, 19].

Salt / Component (Composition [mol%])	ϵ_s [-]	ϵ_∞ [-]	W_0 [-]	κ_0 [m^{-1}]	γ_0 [m^{-1}]	γ' [m^{-1}/K]	$\rho(T)$ [kg/m^3]
LiF (100)	9.04	1.93	10.58	1×10^{12}	3.6×10^{-6}	2.4×10^7	$2370 - 0.50T$
NaF (100)	5.07	1.17	8.82	1×10^{12}	$1.49 \times 10^{-6*}$	$1.52 \times 10^6*$	$2700 - 0.59T$
KF (100)	5.5	1.85	8.27	1×10^{12}	$3.08 \times 10^{-7*}$	$5.28 \times 10^7*$	$2640 - 0.66T$
NaCl (100)	5.90	2.33	6.33	1×10^{12}	5.73×10^{-6}	2.22×10^7	$2140 - 0.54T$
FLiNaK (46.5-11.5-42)	—	—	9.64 [†]	1×10^{12}	—	—	$2680 - 0.685T$
LiF	9.04	1.93	—	—	2.16×10^{-6}	1.25×10^7	—
NaF	5.07	1.74	—	—	1.49×10^{-6}	1.52×10^6	—
KF	5.50	1.85	—	—	3.08×10^{-7}	5.28×10^7	—
FLiBe (66-34)	—	—	7.5 [‡]	1×10^{12}	—	—	$2410 - 0.488T$
LiF	9.04	1.93	—	—	5.69×10^{-1}	3.64×10^4	—
BeF ₂	5.07 [‡]	1.74 [‡]	—	—	3.15×10^{-5}	1.77×10^8	—

* Obtained from calibrated value of corresponding compound in FLiNaK model.

[†] Weighted average of Urbach factors of compounds [14].

[‡] Estimated value used from range of values supplied for alkali halides [14].

strongly disagree with the sharp rise predicted by the model around $\lambda = 20\mu m$. At wavelengths shorter than the experimental data range, where the data seems to predict a steeper decline toward 0, the higher predicted κ_λ of the model would cause significant deviation in absorptive contributions where the spectrum overlaps with the blackbody emission curve.

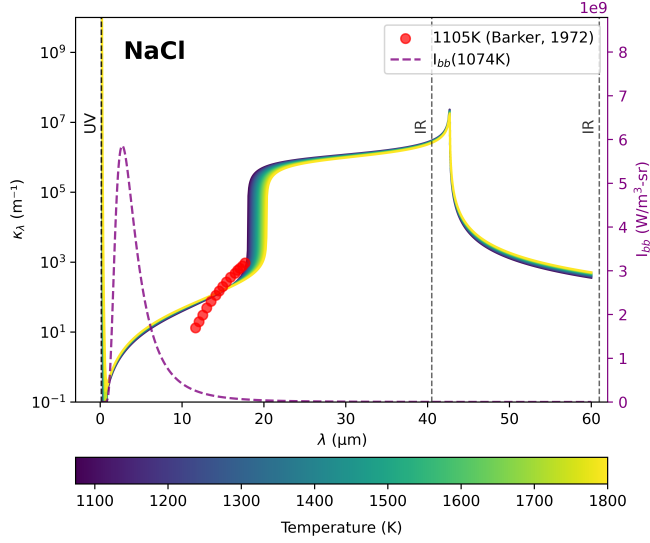
The absorption coefficients for unary salts LiF, NaF, and KF are modeled and plotted in Fig. 3. LiF in Fig. 3a provides a solution fit to experimental data similar in performance to that of NaCl. The principal predicted peak lies in between the first two dominant IR absorption peaks of the solid crystal, which likely corresponds to the elongated bond captured by the $\lambda_{IR} \approx 20\mu m$ peak. The second peak is not informed by the PDF-derived PMF, and is not explicitly accounted for in the model (though thermal radiative absorption at these wavelengths will be minor at the relevant temperatures). The small spike at low frequencies (relative to the other modeled salts) points to the absence of a UV peak in LiF, suggesting that the Urbach parameter may be an overestimation for predicting a peak at all in LiF. Fig. 3b shows the wavelength of the modeled peak that underestimates the measured peak position, indicating a possible contraction of the Na-F bond in molten state or inaccurate force constant-PMF connection. The peak for KF in Fig. 3c matches well with the measured IR peak position, though as noted by Chaleff [3], the high-wavelength KF absorption modes likely contribute negligibly to absorption in a mixture such as FLiNaK. These observations of the model performance for the unary salt provides a reference for the mixture, as discussed below.

The predicted absorption coefficient for FLiNaK is given in Fig. 4. Depicted for comparison is the absorption coefficient calculated by Chaleff et al., which is linearly interpolated from the unary endmember properties, weighted by

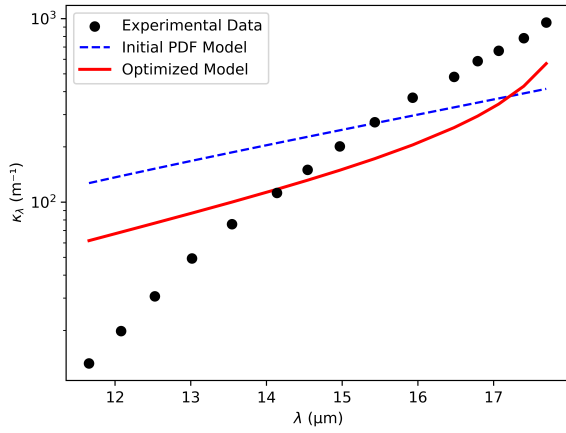
concentration [3]. The absorption correlation given by Chaleff follows a similar shape to the predicted curve, but the locations due not match well. Whereas the endmember interpolation approaches a steeper incline toward 0 at lower wavelengths, the predicted curve immediately increases from near $\lambda \approx 0\mu m$. The sharp peak on the left is the electronic contribution from the Urbach Rule. The first and second peaks in the IR region correspond to the LiF and NaF contributions ($\kappa_{IR,LiF} = 23\mu m$ and $\kappa_{IR,NaF} = 48\mu m$), whereas the peak for KF is beyond the plotted range to $\kappa_{IR,NaF} = 66\mu m$. These exceed the positions of the experimental IR absorption peaks of the unary solid salts, as well as the peak positions for the corresponding endmembers in Fig. 3. This may be explained by the change in fundamental oscillator frequency of each pair depending on the charged ionic environment, where complex charge screening behavior can significantly alter both the pair separation distance and resulting bond strength.

The temperature dependence of the spectrum is also noted. As the temperature increases, the peak flattens and shifts slightly to the right, indicative of red-shifting absorptivity due to the increased thermal energy and decreased negative potential between cation-anion pairs. Interestingly, while the peak positions decrease in the absorption coefficient with increasing temperature, the valleys increase, possibly signaling the more closely spaced energy levels at higher temperatures.

To qualitatively assess the relative change in fundamental frequency due to PMF differences between the unary and the mixture, Fig. 5 depicts the partial PDFs of the Li-F and K-F cation-anion pairs as contained in unary LiF, NaF, FLiNaK, and FLiBe. Not only do the peak positions differ between the mixture context of the cation-anion pair, but the depth of the first minima also varies. More specifically, the ratio of the



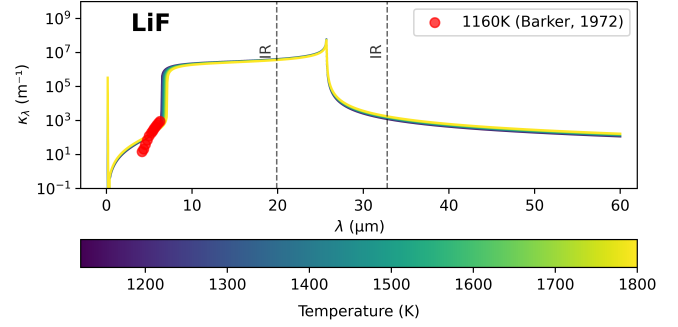
(a)



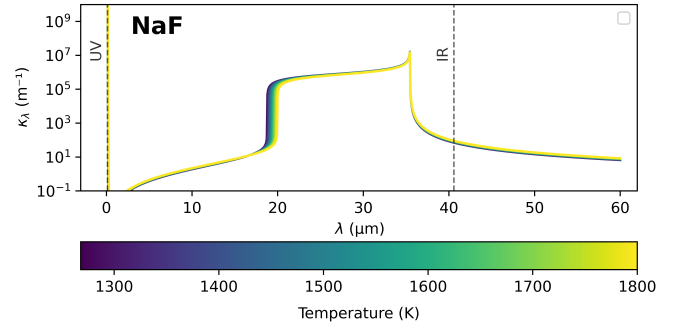
(b)

Fig. 2: (a) The absorption coefficient of NaCl as a function of wavelength and temperature, with the superimposed blackbody Planck distribution (calculated at the temperature at which the PDF was obtained). Prominent UV and IR absorption peaks of the solid salt crystal at room temperature noted by labeled vertical lines. The experimental absorption coefficient is plotted for reference [11]. (b) Plot of the optimized model curve fitted to the experimental data, compared to the initial model.

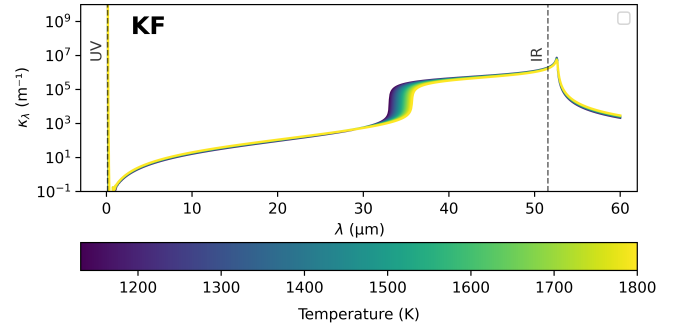
peak height to subsequent minima differs, which is related to the bond strength of the pair; stronger bonds would allow fewer escaping neighbors and a resulting lower first minima. The unweighted PDFs are normalized to 1, which does not reflect their concentrations, but allows for visualization of the qualitative change in pair interactions given different mixture compositions. This highlights the importance of using the mixture PDF to extract the relevant PMF for calculation of the absorption coefficient.



(a)



(b)



(c)

Fig. 3: The absorption coefficient of as a function of wavelength and temperature for (a) LiF compared to experimental data [11]; (b) NaF; (c) KF.

While the parameters of BeF_2 are unavailable, the absorption coefficient of FLiBe can be estimated using the order-of-magnitude parameters noted in Table I and fitting the curve to the experimental absorption spectrum. Fig. 6a provides the plot of κ_λ for FLiBe , which shows a relatively close fit to the experimental data proposed by Liu et al. [?]. The first and second principal peak positions caused by Be-F and Li-F (around $\kappa_\lambda = 7\mu\text{m}$ and $22\mu\text{m}$, respectively) correspond to just past the respective IR absorption peaks of each pair's respective salt crystal.

The Planck-mean wavelength-weighted value for $\kappa_P(T)$ was evaluated across 0–1000 μm for FLiBe across the working temperature range, plotted in Fig. 6b. To demonstrate the connection to practical implementation of the predicted absorption coefficients, $\kappa_P(T)$ was then used for RHT calcu-

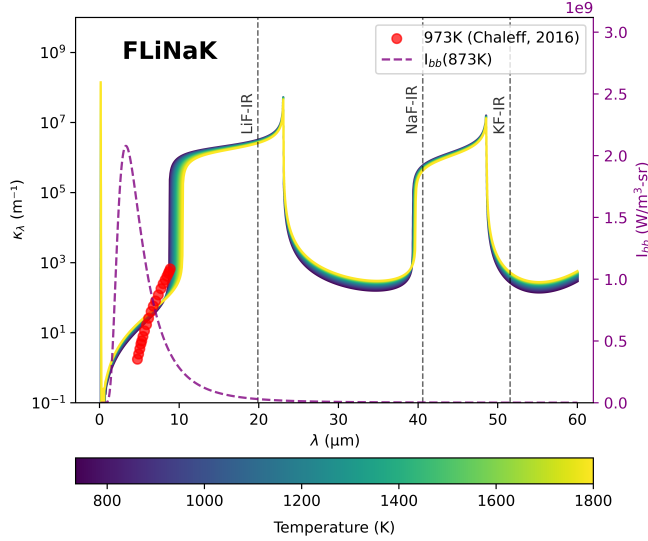


Fig. 4: The absorption coefficient of FLiNaK as a function of temperature and wavelength. The endmember interpolation is from molar averaging predictions proposed by Chaleff et al. [3].

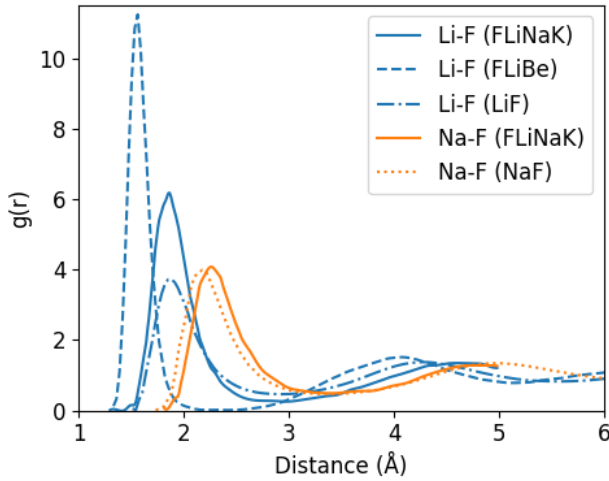
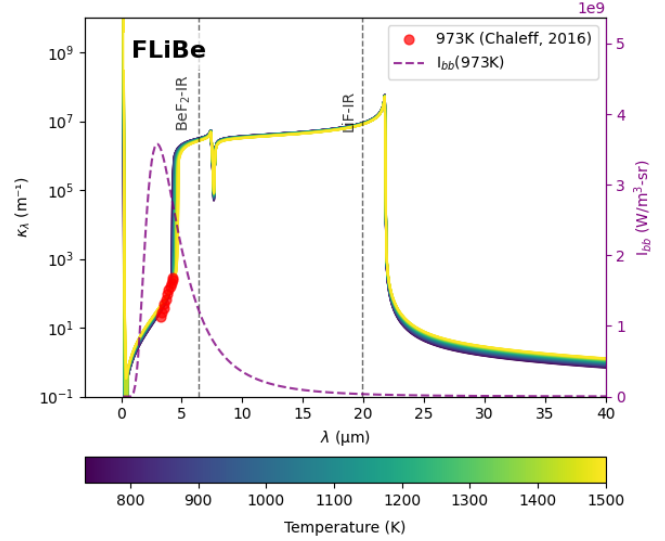


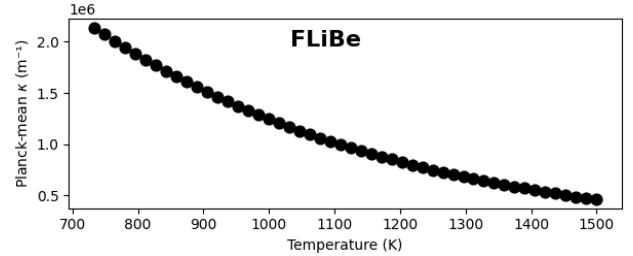
Fig. 5: The unweighted pair distribution functions (PDF) of the respective Li-F and Na-F pairs in unary LiF and NaF and FLiNaK and FLiBe. PDFs taken from [15, 20, 21].

lations in Fig. ?? of Section ??.

Several notable limitations of the model offer mechanistic insights and opportunity for further development. As noted by Derdeyn et. al., [26], impurities beyond simple end-member constituencies can have a drastic effect on the absorptive properties of molten salts. This predictive model does not consider alternative vibrational modes, though it does provide a valuable starting point that directly connects the structure of the molten salt (using the PMF from the PDF) to inform a dominant absorptive character of the salt. Significantly, the peak positions of generated by the PMF provide theoretically reasonable locations compared to the IR absorp-



(a)



(b)

Fig. 6: (a) The absorption coefficient of FLiBe as a function of wavelength and temperature, with the superimposed blackbody Planck distribution (calculated at the temperature at which the PDF was obtained). Prominent UV and IR absorption peaks of the solid salt crystal at room temperature noted by labeled vertical lines. The experimental absorption coefficient is plotted for reference [10]. (b) Plot of the Planck mean absorption coefficient at various temperatures.

tion peaks of the solid compound, suggesting the dominant role of the cation-anion pair in photon absorption in infrared regions. The magnitude of the absorption spectrum is less informed, in large part due to the lack of experimental data. Measuring the absorption coefficient across a broader wavelength range would provide valuable reference points to verify the shape of the spectrum.

The model of absorption coefficients provides a first-order estimate that gives a qualitative connection to molecular structure. Variations in the absorption spectrum can have significant implications for heat transfer modeling, as discussed in Section ?? and summarized in Table ??, where the increased κ_P predicted by the present model caused significant radiative contributions according to the simple model. Uncertainty in supplied model parameters may also influence the accuracy of the model as interpreted. It is reasonable that the fundamental wavelength increases with the mass disparity of the ion pair according to Eq. (). Despite the assumptions

made in the model, the overall position of the spectral absorption coefficient features appear to reflect the contributions of the dominant cation-anion pairs, although the magnitude and other vibrational modes require increased information to incorporate.

Conclusions

The theoretical model of the absorption spectrum provides a direct connection between molecular structure and optical properties. To the authors' knowledge, this is the first time the pair distribution function (PDF) has been used to inform the fundamental frequency for use in a Lorentz oscillator model composed of cation-anion pair contributions. The qualitative similarities in predicted absorption peaks and measured IR peaks in the solid state indicate tentative agreement in the ability of the potential of mean force (PMF) obtained from the second derivative of the PDF to calculate the position of the end-member peaks in the molten salt. The magnitude of the spectra remain less informed, and significant differences exist in the shape of experimentally measured regions. Further measurements across a wider range would provide valuable information to improve the accuracy and mechanistic interpretation of the model.

The discussion of the theoretical model in this work provide an initial approach to connecting first principles mechanics to radiative heat transfer. This work emphasizes the importance of collecting spectroscopic data, both to validate models and provide theoretical interpretation of large-scale heat transfer applications.

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